Agnibh Dasgupta

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Education

University of Nebraska Omaha, Ph.D., Computing & Information Science *2021 – 2026*
Utah State University, M.S., Computer Science *2018 – 2020*

Skills

Research:	Representation Learning, Watermarking, Computer Vision, NLP, SSL, Multimodality
Models:	Transformers, ViT, CLIP, LLMs, Diffusion Models
ML Stack:	Python, PyTorch, TensorFlow, Hugging Face Transformers, CUDA
Training:	LoRA, SFT, DPO, GRPO, RLHF, RLAIF, PyTorch DDP, DeepSpeed, FSDP, Slurm/HPC
Agentic AI:	RAG, VectorDB, LangGraph, Tool Use, Function Calling
Evaluation:	LLM-as-a-Judge, RAGAS, A/B Testing, Human-in-the-Loop
Deployment:	vLLM, Quantization, Cloud Deployment, Docker

Professional Experience

University of Nebraska Omaha & University of Nebraska Medical Center, Postdoc 2026 – Present

- Developing multimodal deep learning methods using video, audio, and speech transcripts to detect and classify challenging behaviors of individuals with disabilities at the Munroe-Meyer Institute.

University of Nebraska Omaha, Graduate Researcher 2021 – 2026

- Led papers and experiments, hiring, interviewing, and mentoring interns in weekly meetings
- Worked in interdisciplinary teams with education researchers at UNL and doctors at Stanford
- Served as a peer reviewer, presented at conferences, authored NSF and graduate grant proposals

Awards

CVPR Compute Transparency Champion Award 2026

- Awarded for exemplary reporting of computational resources, methodology, and reproducibility details

Selected Publications and Projects

Siosa's Library: Path of Exile Q&A Agent *2026*

- Built a **RAG** based tool using **agentic AI** workflows to help new players navigate the game's difficulty
- Designed a **multi-agent** workflow using **LangGraph** to coordinate and tool selection across multiple data sources
- Hosted a pretrained **LLM** on **cloud infrastructure**, optimizing inference with **vLLM** and **quantization**
- Aligned model behavior via **human-in-the-loop post-training** (**DPO**, **PPO**, **GRPO**), evaluated with **LLM-as-judge**

Invariant Features in LLMs: Geometric Characterization and Model Attribution *NeurIPS 2026*

- Investigated whether **LLMs** encode causally meaningful **representations** that can be used for **model attribution**
- Identified **invariant subspaces** in 9 open-source **LLMs** (7-10B params) via contrastive sensitivity decomposition
- Validated semantic and nuisance components causally through next-token prediction interventions
- Achieved 90%+ **zero-shot model attribution** accuracy across base, **fine-tuned**, and **distilled LLMs**

TIACam: Text-Anchored Invariant Feature Learning with Auto-Augmentation *CVPR 2026*

- Addressed the challenge of maintaining **watermark** robustness under real-world camera capture conditions
- Built a **zero-watermarking** framework combining **CLIP** image–text features with a learned **augmentation** module
- Trained a differentiable auto-augmentor to learn camera-like distortions, replacing hand-designed **augmentation**
- Achieved 95%+ watermark recovery on **ImageNet** and **Visual Genome** under real-world camera conditions

Multimodal Sentiment Analysis in K–12 Classrooms *(in progress)* *2026*

- Extending a published **video**-only classroom mood classification pipeline (IEEE FIE 2024) lacking audio
- Collaborating with education researchers at **UNL** to build a **multimodal** system for richer instructor feedback
- Combined **video** and **audio** features (**log-mel spectrograms**, acoustic uncertainty) to model student engagement
- Ongoing: evaluating whether the added audio modality improves on the published video-only baseline

Watermarking Language Models through Language Models *IEEE TAI 2025*

- Addressed the need for **LLM watermarking** that generalizes across architectures without modifying model weights
- Built a model-agnostic, prompt-guided **watermarking** scheme for **LLM**-generated text
- Tested on 25 open-source **LLMs** including **fine-tuned** and **distilled** variants using **LoRA** and **DeepSpeed**
- Achieved 88%+ detection accuracy under paraphrasing, **fine-tuning**, and **adversarial prompt attacks**

Robust Image Watermarking based on Cross-Attention and Invariant Domain Learning *CSCI 2023*

- Addressed the need of a **watermarking** scheme robust to real-world noise via a learned representation
- Built a **ViT**-based **cross-attention watermarking** system with a **self-supervised** triplet loss objective

- Trained the encoder to cluster augmented views while separating different images in latent space
- Achieved 90%+ **bit recovery rate** (**BRR**), comparable to prior state-of-the-art under compound noise

Perspective Transformation Layer *CSCI 2022*

- Addressed the limitation of fixed preprocessing pipelines for handling viewpoint variation in image recognition
- Proposed a lightweight learnable layer for perspective-robust recognition, replacing fixed pipelines
- Evaluated on **SVHN**, Imagenette, and **MNIST** under viewpoint variation with minimal extra compute
- Achieved 96%+ classification accuracy, outperforming prior state-of-the-art on perspective benchmarks

Deep Q-Learning Applied to Stock Trading *M.S. Thesis, 2020*

- Designed a **reinforcement learning** agent to **predict** next-day **S&P 500** price movement from historical index data
- Represented price history as **candlestick charts** to emulate human trader pattern recognition
- Automated daily scraping, **prediction**, and **model** fine-tuning into one self-updating pipeline
- Observed positive portfolio returns from the agent's **prescribed** buy/sell/hold actions